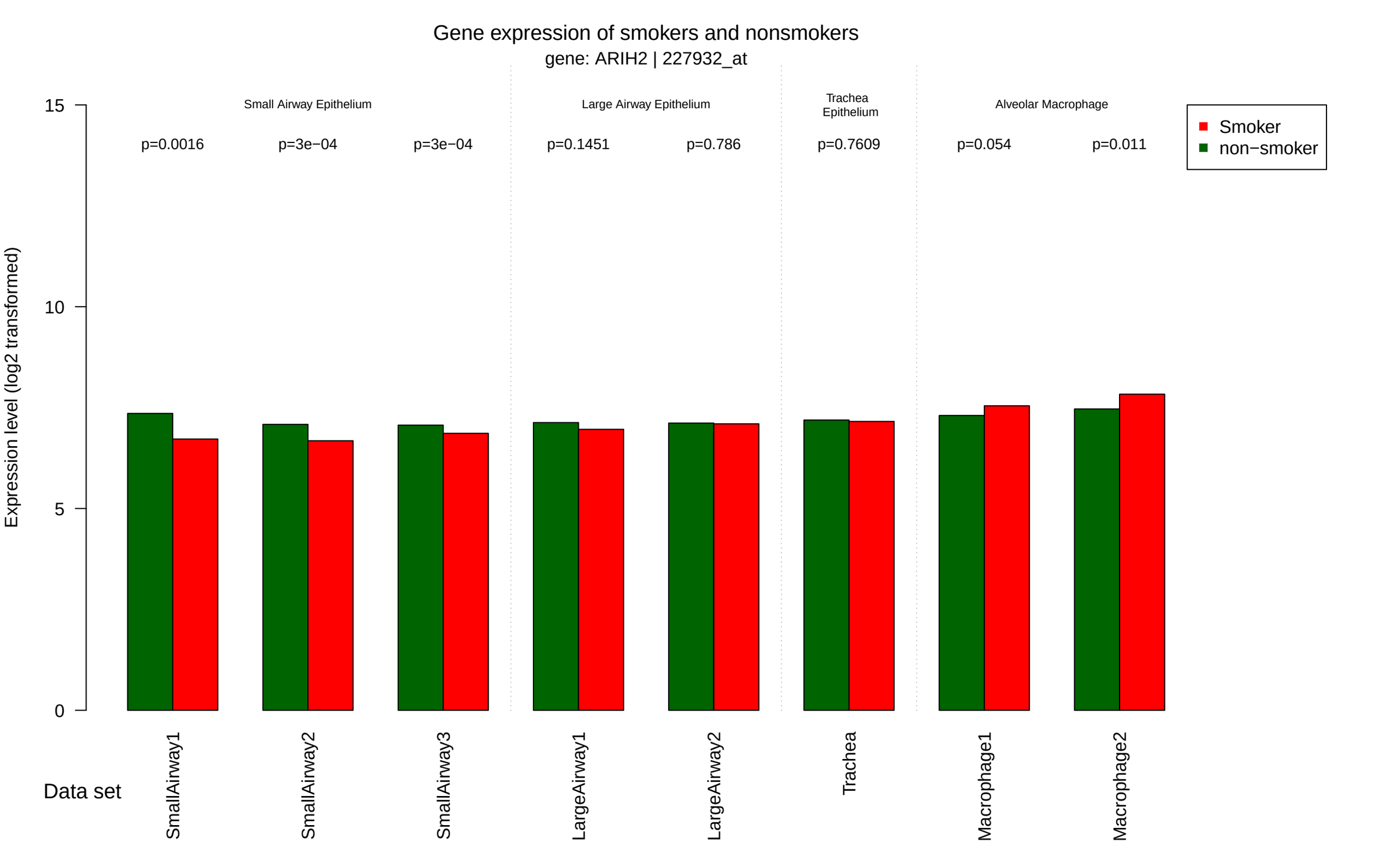
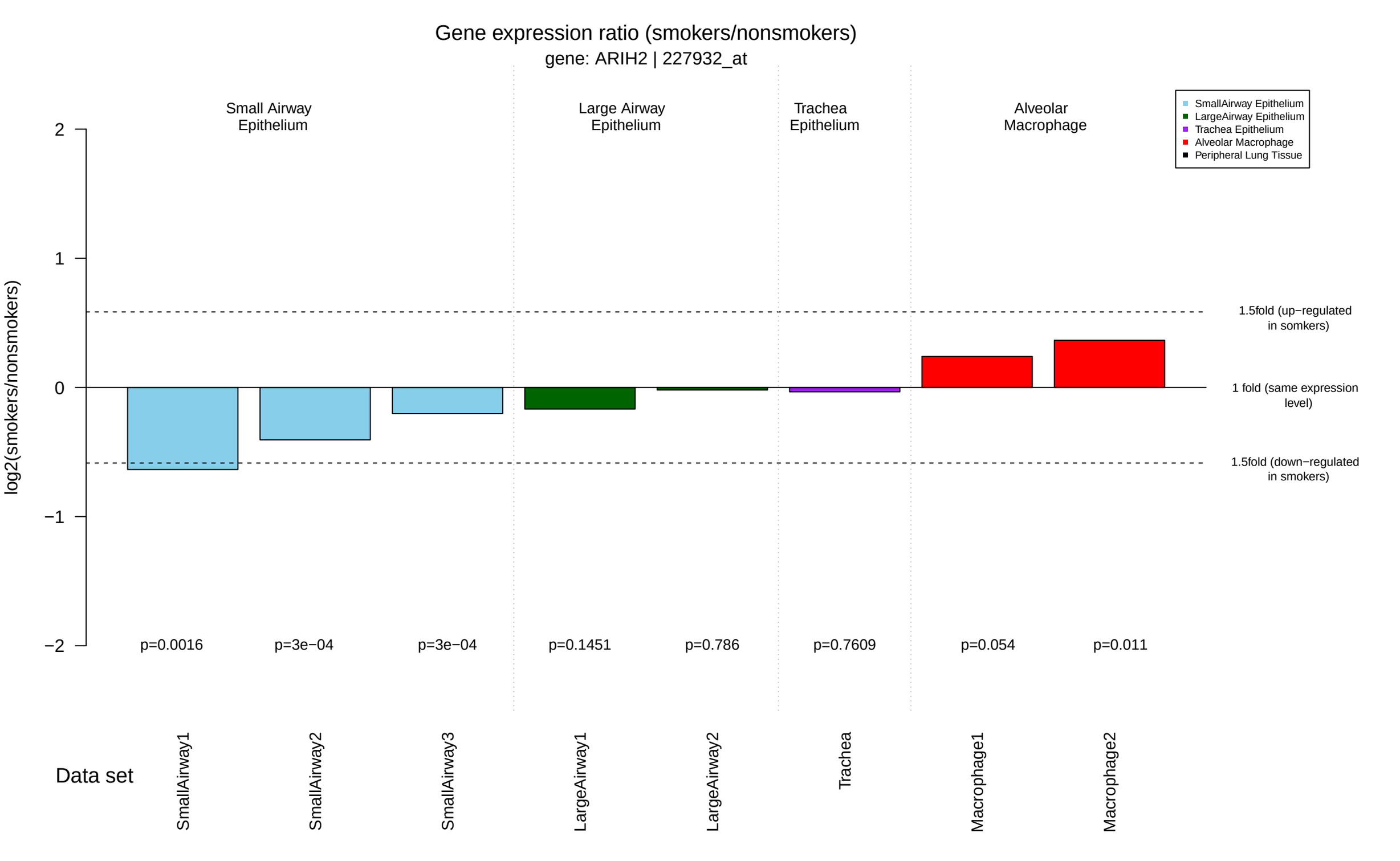
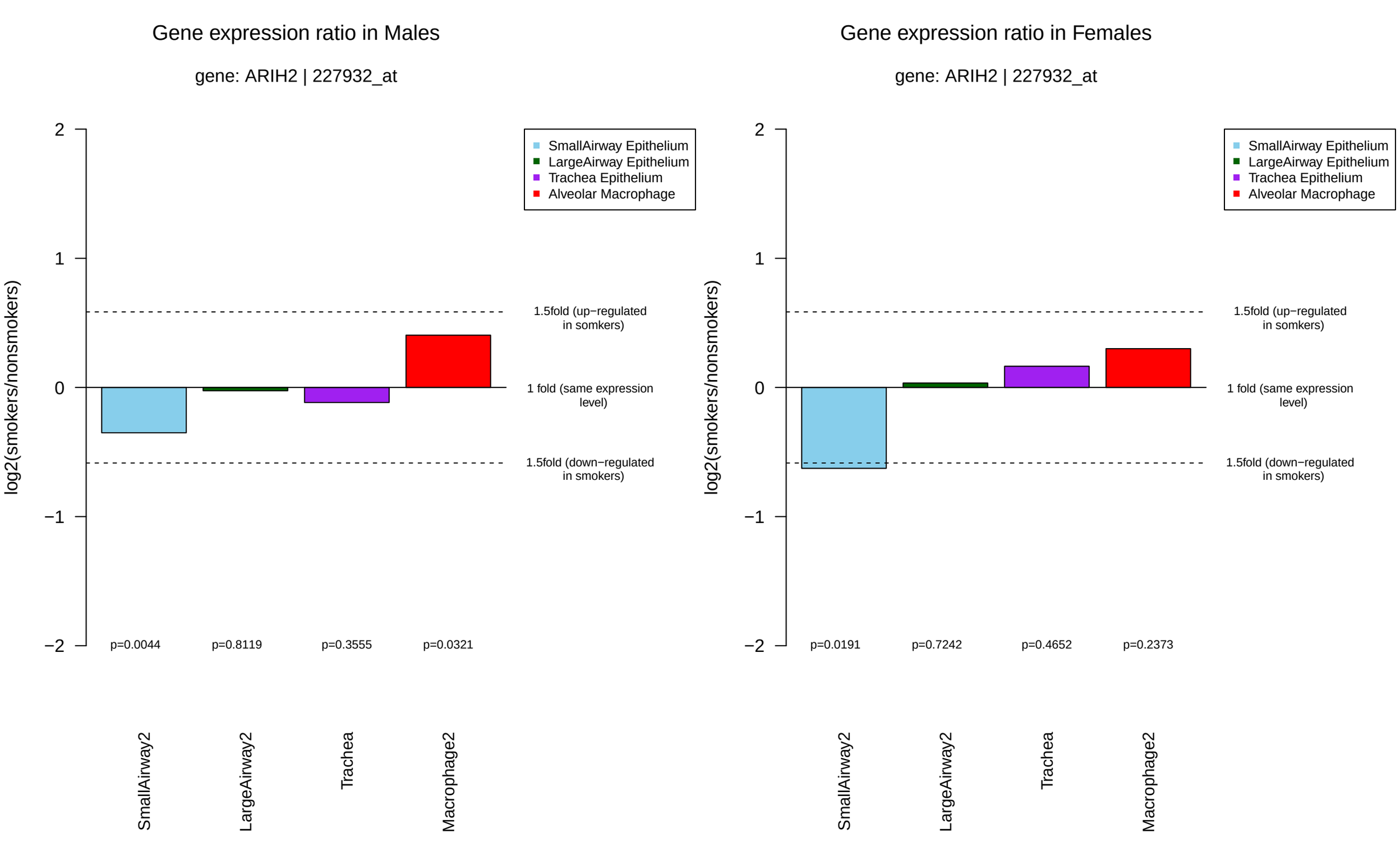
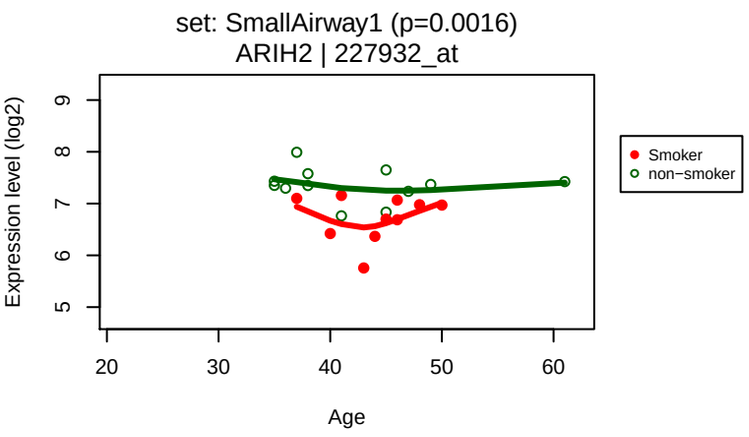
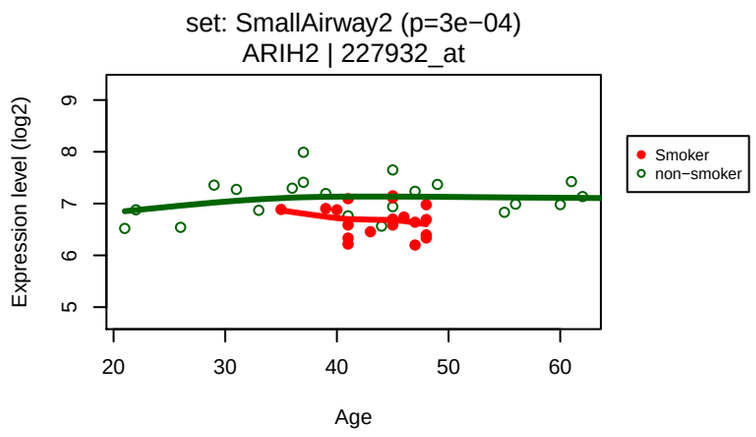
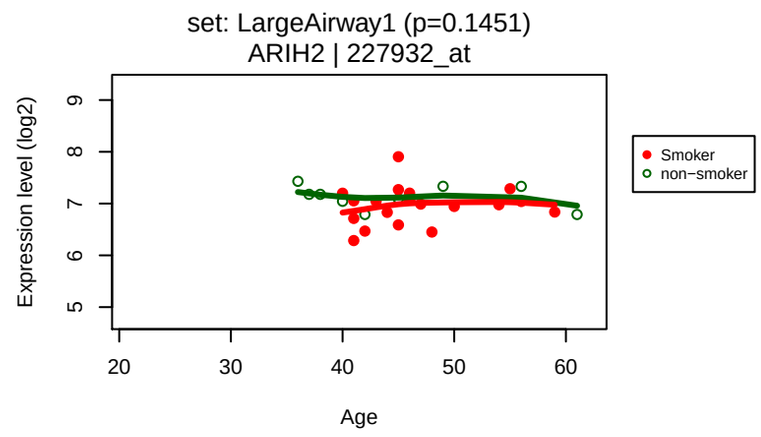

Gene expression of smokers and nonsmokers
gene: ARIH2 | 227932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0016
p=3e−04
p=3e−04
p=0.1451
p=0.786
p=0.7609
p=0.054
p=0.011
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

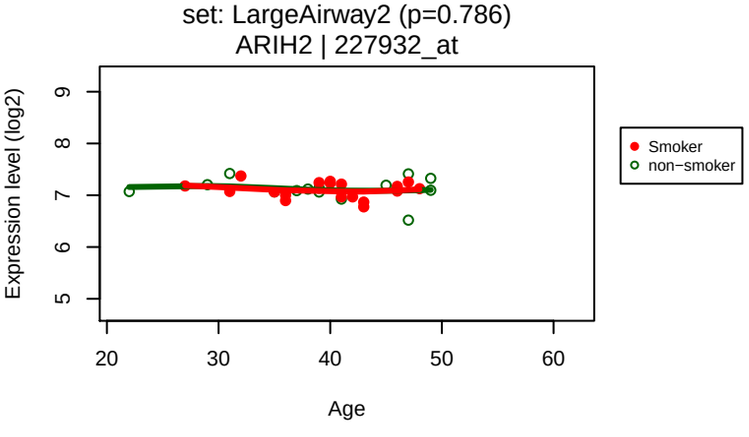

Gene expression ratio (smokers/nonsmokers)
gene: ARIH2 | 227932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=3e−04
p=3e−04
p=0.1451
p=0.786
p=0.7609
p=0.054
p=0.011
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

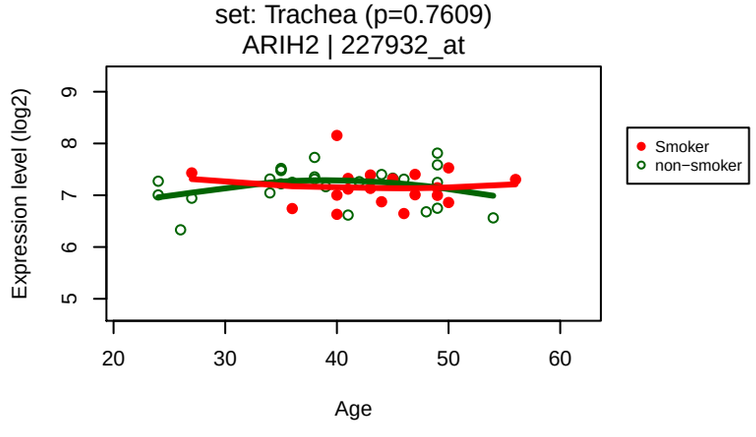

Gene expression ratio in Males
gene: ARIH2 | 227932_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.8119
p=0.3555
p=0.0321
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARIH2 | 227932_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0191
p=0.7242
p=0.4652
p=0.2373
SmallAirway2
LargeAirway2
Trachea
Macrophage2

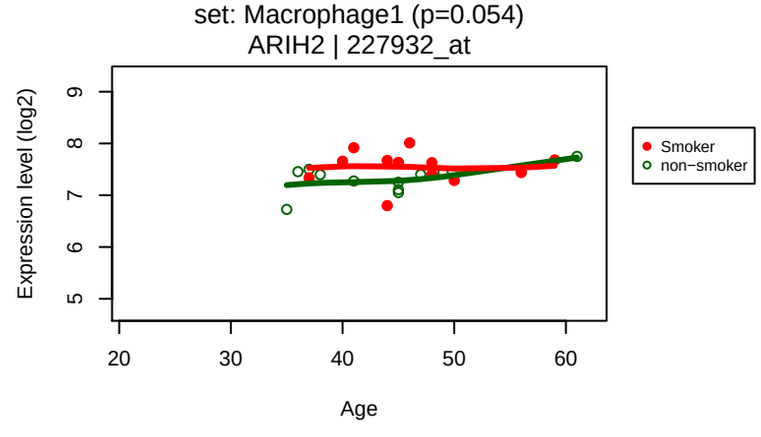

set: SmallAirway1 (p=0.0016)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker

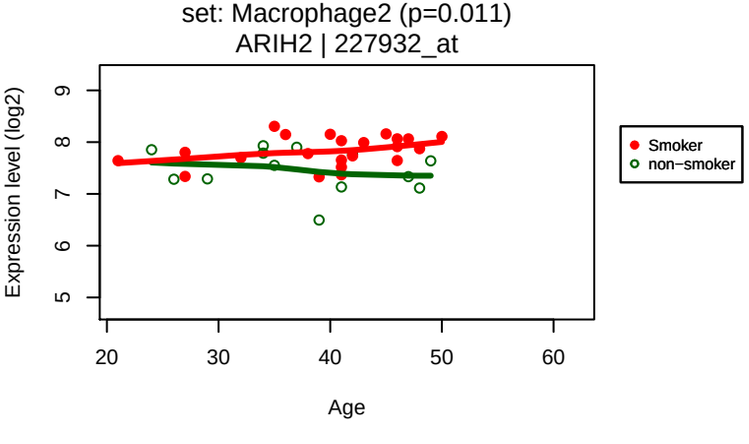

set: SmallAirway2 (p=3e−04)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker

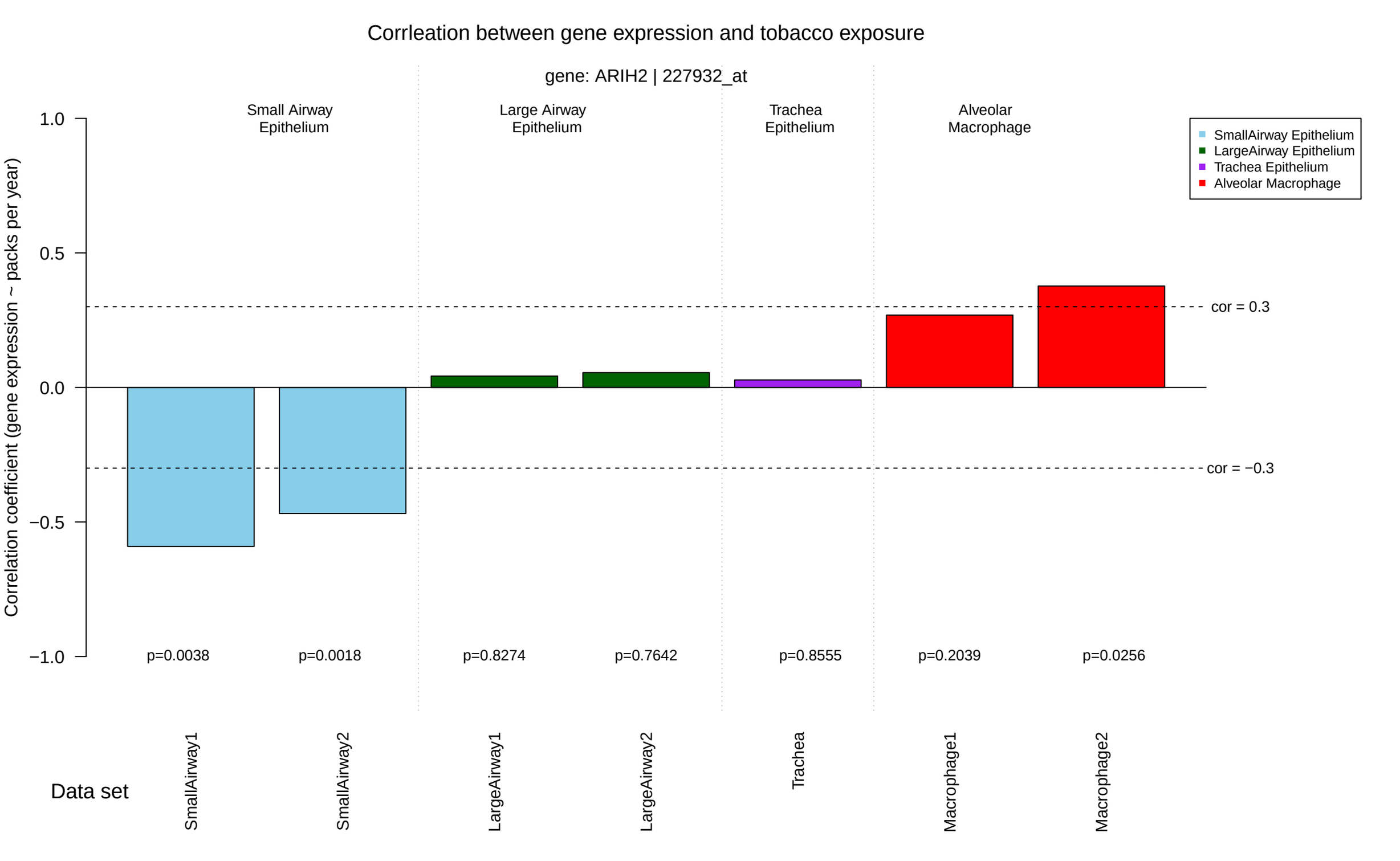

set: LargeAirway1 (p=0.1451)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.786)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7609)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.054)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.011)
ARIH2 | 227932_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARIH2 | 227932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0038
p=0.0018
p=0.8274
p=0.7642
p=0.8555
p=0.2039
p=0.0256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2